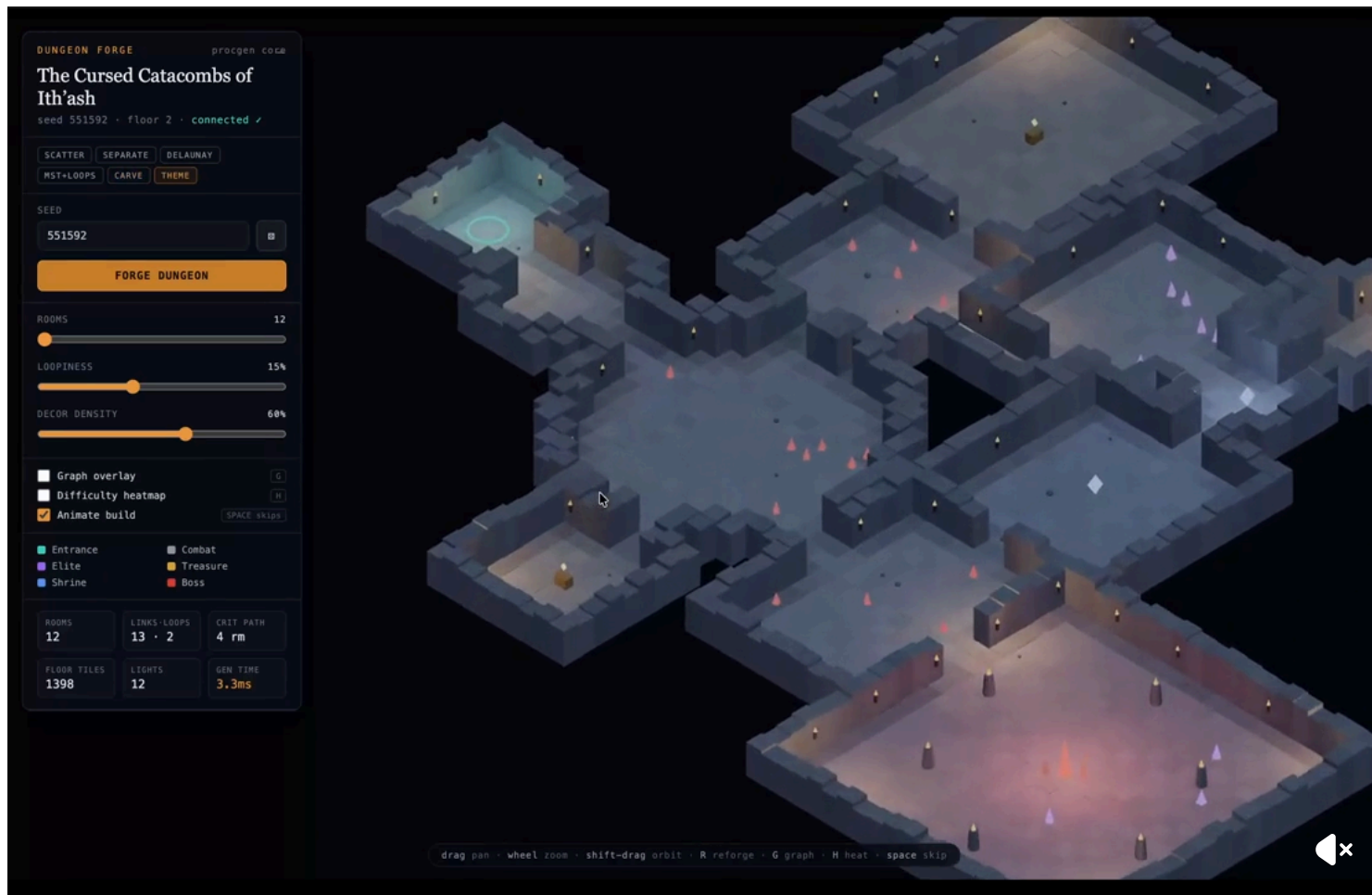


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r/aigamedev • 22h ago
MajidManzarpour

Real-time procedural dungeon generator in three.js, built with Fable 5

[Demo](#) | [Project](#) | [Workflow](#)

Shared the prompt in this thread if you want to incorporate into your game

<https://x.com/majidmanzarpour/status/2073742439211225234>

Update, open sourced: <https://github.com/majidmanzarpour/threejs-procedural-dungeon>

535

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adobe • Promoted

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Search Comments

**Sea-River4818** • 22h ago

I've never actually looked into it, is three.js a game engine or coding language? And how does it look so good?

15 Reply ...

**MajidManzarpour** OP • 21h ago

It's a 3D engine, you can build custom game logic on top of it. I put out an open source skill pack for coding agents that might help you get started <https://github.com/majidmanzarpour/threejs-game-skills>

26 Reply ...

**Sea-River4818** • 21h ago

Thanks!

**sooki10** • 18h ago

Nice one brotha

**GoodWillPower** • 17h ago

Thank you!

**BillionBouncyBalls** • 13h ago

This is super cool and inspires me to make a shareable git of the skills I've been building and also inspires me to make a game! 🙌

**Morph_Games** • 18h ago

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**Nez_Coupe** • 11h ago

After enough additions you will have built a game engine however.

**Nez_Coupe** • 11h ago

I had the same question. Ask GPT about it, it's kinda fantastic and very easy for LLMs to understand.

**WORKABLE** • 20h ago

The detail of adding this animation is exactly how we turn pure AI slop into cool things, well done!

**YYY003003** • 21h ago

Yeah. Im thinking its sweet as fuck.

**superkickstart** • 21h ago

Really cool visuals. I did similar one once and it took a week to make. Now it's just hours.

**Morph_Games** • 18h ago

just hours

...and money.

**Iggyhopper** • 14h ago

And two weeks to read the generated code if you want to make a change.

**kauthonk** • 19h ago

yeah, it took me a long time too and didn't come out nearly so clean - going to try and rebuild this month and see what's possible.

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meshy.ai

**rodriguezcappsec** • 17h ago

I am experimenting to see if I can turn the prompt that instead of 3d its 2d top down and instead of three.js, Godot or Phaser.js

**rodriguezcappsec** • 15h ago

Also doing another experiment where I can generate no just dungeons, but maps, open world maps, cities, inside of places like houses, taverns, castles etc etc.

**MajidManzarpour** **OP** • 17h ago

Should be possible! Let us know how it goes

**rodriguezcappsec** • 16h ago

It's working flawlessly!!! will upload video in a bit.

**Leehammy** • 1h ago

Amazing. Is the prompt you sent how you converted it to 2d? You got this in a public GitHub?

**rodriguezcappsec** • 17h ago

Definitely. On it!

**MajidManzarpour** **OP** • 16h ago

Open sourced btw <https://github.com/majidmanzarpour/threejs-procedural-dungeon>

**rodriguezcappsec** • 15h ago

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ROLE

You are a senior gameplay engineer specializing in procedural content generation and real-time 2D rendering in Godot. You write deterministic, allocation-conscious, statically typed GDScript and understand TileMapLayer batching, MultiMesh2D instancing, canvas-item draw-call behavior, and 2D lighting costs at a deep level. You ship code that self-verifies.

CONTEXT

Target: a 2D top-down, Diablo-style ARPG built in **Godot 4.4+** (GDScript, no addons, no C# build step). Camera is a plain top-down `Camera2D` with zoom + drag-pan. World scale: **1 grid cell = 1 tile = 16 px** (expose `TILE_SIZE` as a constant), grid `(x, y)` maps directly to TileMap cell `(x, y)`; world position = `cell * TILE_SIZE`. Generation runs on the main thread on level load. Everything lives in a minimal project: one scene, three scripts.

OBJECTIVE

Implement `generate_dungeon(params) -> Dungeon` (pure data, zero Node/scene-tree/server dependencies — `RefCounted` only) plus `build_dungeon_scene(dungeon) -> Node2D` (pure presentation). Dungeons must have:

- **Character** — irregular silhouettes, mixed room shapes, themed set-pieces, torch-lit atmosphere, a seeded name.
- **Depth** — branching with dead-end rewards, guaranteed loops (never a pure tree), secrets off the critical path, tension/release pacing.
- **Challenge** — a difficulty ramp keyed to graph distance from the entrance, elite gates on the critical path, boss at maximal depth.

NON-NEGOTIABLE CONSTRAINTS

1. **Determinism.** One integer seed reproduces the dungeon bit-for-bit, across engine versions. Global `randi()`/`randf()`, `randomize()`, `Time.get_*` feeding logic, and iteration order over unordered collections feeding logic are banned. Implement **mulberry32** in GDScript (mask every step with `& 0xFFFFFFFF` — GDScript ints are 64-bit) and thread the RNG object explicitly through every stage. Do not rely on `RandomNumberGenerator` (engine-version coupling).
2. **Connectivity.** Flood fill from the entrance must reach 100% of floor cells. On failure, re-roll internally with a derived seed (max 5 attempts) — never ship a broken layout, never "fix" islands with teleporters.
3. **Performance.** ≤ 50 ms generation at 60 rooms (measure with `Time.get_ticks_usec()`). Level geometry rendered by **TileMapLayer** nodes (one floor layer, one wall layer) plus **one** `MultiMeshInstance2D` **per prop kind** — zero per-tile `Sprite2D`/`Node2D` instances. ≤ 12 `PointLight2D`s, `shadow_enabled = false` on all of them.

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releases textures on regenerate.

PIPELINE — implement as discrete, individually testable stages

1. **RNG.** mulberry32 wrapped in a small class with helpers: `float_range(a, b)`, `int_range(a, b)`, `pick(arr)`, `chance(p)`, `gaussian(mu, sigma)` (Box-Muller).
2. **Room scatter.** Spawn `room_count * 1.4` candidates inside an ellipse (radius $\propto \sqrt{\text{room_count}}$ so density stays constant). Archetype table — small 5–7 (45%), medium 8–12 (40%), large 13–18 (15%). Shape table — rectangle 60%, ellipse 22%, chamfered octagon 18%. Force ≥ 2 large rooms.
3. **Separation.** Iterative AABB push-apart with 2-cell padding until stable (cap 300 iterations), snap centers to integer grid, cull the smallest overflow down to `room_count`.
4. **Connectivity graph.** Bowyer-Watson Delaunay over room centers $\rightarrow$ Prim MST as the skeleton $\rightarrow$ re-add non-MST Delaunay edges with probability `loop_chance` (default 0.15), rejecting edges longer than $2.2\times$ the mean MST edge. Loops are mandatory for ARPG flow: report the cyclomatic number ($E - V + 1$) and require ≥ 1 at defaults.
5. **Semantics before carving.** Boss = largest-area room. Entrance = degree-1 room maximizing graph distance from the boss. Critical path = BFS entrance $\rightarrow$ boss; remaining leaves $\rightarrow$ treasure (cap 4); 1–2 shrines mid-depth off-path; 1–2 elite arenas on the critical path at 55–85% depth; everything else combat. `difficulty = 0.15 + 0.85 * (depth / max_depth)`, boss = 1.0.
6. **Corridor carve.** L-corridors center-to-center with seeded elbow direction (skip the elbow when spans overlap enough for a straight run). Width 3 on the critical path, 2 default, 1 permitted on treasure spurs. Stamp FLOOR; tag corridor cells for styling and doorway detection.
7. **Rasterize.** `PackedByteArray` grid of {VOID, FLOOR, WALL}; WALL = any VOID cell with an 8-neighbor FLOOR (render only these — outer void stays empty and vanishes into darkness). Doorway = corridor cell 4-adjacent to room floor. Compute a per-cell BFS distance field from the entrance (`PackedInt32Array`, -1 for non-floor) — reused for pacing, reveal animation, and validation.
8. **Decoration (data only).** Pillar grids in large rooms (only cells whose 8 neighbors are all floor, ≥ 2 cells from any doorway). Torches on floor-facing walls with min Chebyshev spacing 4. Debris density $\propto$ `decor_density`, higher in low-difficulty rooms. Braziers ringing the boss arena; one chest per treasure room; shrine crystal; entrance portal ring. Enemy spawns: `round(area / 18.0 * (0.5 + difficulty))` for combat/elite, none in entrance/treasure/shrine, never on a prop or doorway cell.
9. **Presentation metadata.** Seeded dungeon name from syllable tables ("The Ashen Vaults of Vor'gul"), per-room tint, stats block `{rooms, edges, loops, critical_length, floor_tiles, wall_tiles, props, gen_ms}`.

RENDERING SPEC

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(cracked, broken-top, mossy) selected per cell — this replaces 3D height jitter for a ruined silhouette. Optional terrain/peering-bit autotiling for wall edges.

- **Baked per-cell AO/tint:** render one `Image` at 1 px per cell — floor color = base × (1 − 0.09 × min(adjacent_walls, 4)) ± 5% value noise, blended ~18% toward the room tint; corridors darker and untinted; VOID fully black. Upload as a single full-dungeon `Sprite2D` scaled to `TILE_SIZE`, multiply-blend `CanvasItemMaterial` over the floor layer. One draw call, zero per-tile nodes.
- **Props:** one `MultiMeshInstance2D` per kind — pillar, torch bracket, flame, debris, chest, spawn marker, crystal, brazier — with per-instance transform + color. Flames flicker via per-frame instance color/scale jitter (or a tiny canvas shader driven by `TIME` + per-instance custom data).
- **Lighting:** `CanvasModulate` at a dim near-black blue for global darkness; warm `PointLight2D`s (color `#ff8c3a`, radial-gradient texture, `texture_scale` ≈ 9 tiles, energy flicker per frame) on a **farthest-point-sampled subset** of torches within the 12-light budget, plus key lights at entrance (cool), shrine (teal), boss (red). All lights `shadow_enabled = false`.
- **Depth falloff:** a radial vignette + the AO overlay stand in for fog; optionally reveal floors progressively using the BFS field for a fog-of-war feel.
- **Debug overlays (toggleable):** one `Node2D` using `_draw()` — Delaunay (faint), MST (white), loop edges (cyan), critical path (red), difficulty heatmap recolor of the AO overlay.

DATA CONTRACT

```
class Dungeon extends RefCounted:
  var params: Params
  var name: String
  var w: int
  var h: int
  var grid: PackedByteArray      # VOID | FLOOR | WALL
  var bfs: PackedInt32Array      # per-cell distance from entrance, -1 = non-f
  var rooms: Array[Room]        # { id, cx, cy, w, h, shape, type, depth, dif
  var edges: Array[Edge]        # { a, b, is_loop, is_critical }
  var doorways: Array[Vector2i]
  var corridor_cells: Array[Vector2i]
  var props: Array[Prop]        # { kind, x, y, rot, scale, room_id }
  var spawns: Array[Spawn]      # { x, y, tier, room_id }
  var stats: Stats
```

ACCEPTANCE TESTS — must run automatically on startup and print results

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- boss depth $\geq 60\%$ of max BFS depth; entrance degree = 1; entrance $\neq$ boss-adjacent.
- ≥ 3 leaf rooms at 40+ rooms; loop count = cyclomatic number = $E - V + 1$.
- No prop or spawn on a doorway, wall, or void cell; light count within budget.
- 60-room generation < 50 ms (report the measured value via `Time.get_ticks_usec()`).

TUNABLES (expose with defaults)

```
seed, room_count = 42, loop_chance = 0.15, decor_density = 0.6, theme = "crypt",
TILE_SIZE = 16
```

ANTI-GOALS

Uniform room sizes; corridor spaghetti (> 2 elbows per link); boss reachable in $< 60\%$ of max depth; per-tile `Sprite2D` nodes; shadow-enabled 2D lights; hidden nondeterminism (`randomize()`), unseeded globals, engine RNG coupling); a pure-tree layout with zero cycles.

DELIVERABLE

A minimal, self-contained Godot 4.4+ project:

- `dungeon_generator.gd` — pure data generator, headless-runnable.
- `dungeon_builder.gd` — scene construction + teardown.
- `main.tscn` / `main.gd` — `Camera2D` (zoom + drag-pan) and a control panel built from `Control` nodes: seed `LineEdit`, dice `Button`, `HSlider`s for `room_count` / `loop_chance` / `decor_density`, `CheckBox` overlay toggles, live stats `Label`, legend.

Include an optional staged build animation gated behind `animate_build` (Tweens + the BFS field) — rooms scatter→separate, graph resolves Delaunay→MST+loops, floors flood outward along the BFS distance field, wall variants stamp in, props pop — so the algorithm itself is legible on screen.



sassyhusky • 8h ago

20 y engineer here: this prompt is hot garbage, it reads like a 1y junior trying to sound smart and used bullshit generator to write it, it literally went "generate bs system design words mess vaguely resembling a product, make it as verbose as possible". I recommend Cub_UK you 17h ago don't let these things write specs and just use plain English with as much detail as possible and let it do what it does best (which obviously isn't writing specs). Would love this if it goes well!



Cub_UK you 17h ago

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**MajidManzarpour** OP • 14h ago

Awesome work! Glad it was helpful in your project

**Infinitecontextlabs** • 21h ago

Awesome. Keep going.

**MajidManzarpour** OP • 21h ago

Thanks! 🙏

**AndroidJunky** • 20h ago

This is neat! I've been building a rogue clone as a weekend project since the early days of agentic code completion. I spend days building a level generator and had so much back and forth with ChatGPT at that time, it wasn't even funny anymore. The release of Fable accelerated the development overnight. So much progress you can do in a single 5 hour window.

**u/anthropic-ai** • Promoted

Legora lets lawyers draft, review, and research in a single workspace. Here's how Max Junestrand is building it with Claude.

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"Claude gives us the flexibility to build more diverse solutions faster."

Max Junestrand, Legora co-founder & CEO

**Benhamish-WH-Allen** • 20h ago

It's good now just tell fable to make it fun

**SleepDeprivedDad_** • 20h ago

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**MajidManzarpour** OP • 16h ago

Open sourced if you want to try incorporating in your game <https://github.com/majidmanzarpour/threejs-procedural-dungeon>

**Ambitious-Crew-7132** • 20h ago

Thats awesome, but if i wish it could go endlessly instead

**MajidManzarpour** OP • 18h ago

I think that could be extended as a feature!

**datNovazGG** • 19h ago

Have you tried the same prompt with other (less capable) models and how did that turn out?

Pretty cool I must say!

**MajidManzarpour** OP • 18h ago

Actually yes, going backwards from Fable to prompt other models seems to work well. Gemini was able to produce something similar with the prompt I shared in the thread.

**DowntownAd3538** • 19h ago

Looks good!

**gorliggs** • 19h ago

This is awesome.

**MeAndClaudeMakeHeat** • 19h ago

this is pretty dope :)

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**Smok3dSalmon** • 16h ago

Wow this is very cool. Why not just share the code on github?

**MajidManzarpour** **OP** • 16h ago

Enjoy <https://github.com/majidmanzarpour/threejs-procedural-dungeon>

**Smok3dSalmon** • 16h ago

Thank you!

**MajidManzarpour** **OP** • 17h ago

Repo coming soon

**justsaymoo** • 18h ago

looks so cool! very good instrument for a fastest work

**rodriguezcappsec** • 17h ago

pretty awesome!!

**Toamig** • 17h ago

How much did it cost you in credits?

**MajidManzarpour** **OP** • 16h ago

Not much actually, Fable built out the initial version in ~15 mins. The open source version has more features

**maxhit_studios** • 15h ago

Looks great

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Looks good! Thanks for sharing!



Woltragen • 11h ago

nice



loontoon • 10h ago

That looks really cool.

I wish there was a way I could incorporate that into my game, but my game is a series of rooms with various mechanical puzzles the player needs to solve to get to the next room; and there is no way AI can make the design decisions required to populate a floor with the right assets to make puzzles.



Sea-River4818 • 5h ago

You are a hero making it open source, I was struggling to get a good consistent dungeon layout generation



Outrageous_Band9708 • 18h ago

would love to read the source. but not using x. can you share a github link?

the procedural dungeon generator, not the skill



MajidManzarpour OP • 16h ago

Enjoy <https://github.com/majidmanzarpour/threejs-procedural-dungeon>



MajidManzarpour OP • 18h ago

Repo coming soon



The_Great_Pains • 16h ago

That would be fun to play around with. I'm a fan!

p.s. The spider looks really neat too!

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[\[deleted\]](#) • 37m ago

Is there a content library for Unity?

[curious-of-2020s](#) • 37m ago

Thanks, but what kinda games I can build in Threejs? Only web?

[\[deleted\]](#) • 21h ago

Comment deleted by user

[re-skob](#) • 21h ago

I'd love to see a capable dev reproduce all of this in a few hours - the UI, realtime geometry generation, animation lighting setup, offset wall heights, and all of the little polish items like the expansion effect and animation, and not just the most basic 2d room generator. This is easily a week+ of work from a capable dev and an impossible amount of work from an incapable dev as they don't even notice all of the details as to be able to implement them from scratch.

[CharlesDuck](#) • 20h ago

Yeah that flood fill with floor tiles and wall heights is such a nice touch 🐸

[re-skob](#) • 19h ago

Guy deleted his reply, but he mentioned he was only talking about the "underlying systems and not the pointless embellishments." - this is what I'm talking about when you have an incapable dev who would never be able to replicate this. "Pointless embellishments" take up around 80% of dev work as it's anything but pointless - it's user experience - and it separates successful products from fun little prototypes.

[Forsaken-Leader-1314](#) • 17h ago

Absolutely. Hacking together a prototype with the core functionality is often quick and relatively painless. It's the polish and detail that takes up most of the time.

[lemonlemons](#) • 21h ago

I don't know man if I can get a model that can oneshot all "solved problems", I would be pretty happy.



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IMO "AI can create entire game systems from 1998 with very little human involvement" is still pretty impressive